

## NLP Lab 1

```
PS C:\Users\lokes\nlp lab c> python nlplab1.py
```

```
The boys were running happily
```

```
Tokens:
```

```
['The', 'boys', 'were', 'running', 'happily']
```

```
Stemmed Words:
```

```
['the', 'boy', 'were', 'run', 'happili']
```

```
Lemmatized Words:
```

```
['The', 'boy', 'were', 'running', 'happily']
```

## NLP Lab 2

```
PS C:\Users\lokes\nlp lab c> python nlp1ab2.py
```

```
processing -> NN
```

```
improves -> VBZ
```

```
text -> JJ
```

```
analysis -> NN
```

```
Tag Meanings:
```

```
NN -> Noun
```

```
VB -> Verb
```

```
JJ -> Adjective
```

```
RB -> Adverb
```

```
PRP -> Pronoun
```

```
DT -> Determiner
```

```
Total Words: 6
```

## NLP Lab 3

```
PS C:\Users\lokes> python nlplab3.py
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\lokes\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\lokes\AppData\Roaming\nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
Enter number of headlines: 4
Enter headline 1: Stock market gains record profits
Enter headline 2: Investors celebrate rise in share prices
Enter headline 3: Football team wins championship
Enter headline 4: Cricket tournament attracts audience

Cosine Similarity Matrix:
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]

Headline Clusters:
Stock market gains record profits -> Cluster 0
Investors celebrate rise in share prices -> Cluster 0
Football team wins championship -> Cluster 0
Cricket tournament attracts audience -> Cluster 1

Enter first word: cricket
Enter second word: wins
WordNet Similarity: 0.2
```

## NLP Lab 4

```
PS C:\Users\lokes> python nlp1ab4.py
```

```
TF-IDF Similarity Scores:
```

```
Document 1 : 0.0
```

```
Document 2 : 0.554
```

```
LSA Similarity Scores:
```

```
Document 1 : -0.0
```

```
Document 2 : 0.996
```

```
Most Relevant Document:
```

```
deep learning is used in artificial intelligence
```

## NLP Lab 5

```
PS C:\Users\lokes> python nlp1ab5.py
```

```
ABC -> ENTITY
```

```
Corporation -> ENTITY
```

```
New -> ENTITY
```

```
York -> ENTITY
```

```
Court -> ENTITY
```

```
Enter actual number of entities: 8
```

```
Predicted Entities: 7
```

```
NER Accuracy: 87.5 %
```

## NLP Lab 6

```
PS C:\Users\lokes> python nlp1ab6.py
```

```
Enter biomedical sentence: Aspirin reduces fever in patients
```

```
Actual Relation (1/0): 1
```

```
Tokens:
```

```
['aspirin', 'reduces', 'fever', 'in', 'patients']
```

```
Predicted Relation: 1
```

```
Precision: 1.0
```

```
Recall: 1.0
```

```
F1-Score: 1.0
```